Capacity Meters That Change Sign: Diagnosing Zero-Cost NAS Proxy Transport Across Search Spaces

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Abstract

Zero-cost NAS proxies rank architectures from a single pass at initialization, yet a score that ranks with accuracy in one setting can rank against it in another. Using a 29-feature zoo on the topology and size axes of NAS-Bench-201, we observationally separate a dominant capacity-associated factor from an early-trainability residual. The factor explains $52.9\%/49.4\%$ *of zoo variation and remains associated with effective rank under raw/global orientation, but the cross-axis loading cosine falls from* 0.87 *under outcome-informed orientation to* 0.46*. Several raw proxies also reverse sign, including the condition-number sum* ($+0.88 \rightarrow -0.77$)*; dimension and matched-sample controls support a dimension-linked/proxy-geometry-dependent failure, not an independent mechanism, while matched-sample-count resampling preserves the headline reversals. Two controlled search traps incur* $+5.40$ *and* $+1.97$ *point costs at* $K{=}10$*, while two null directions incur zero. The diagnostic uses cached target-calibration labels and is not fully label-free. These benchmark-bound results motivate validating proxy orientation before transporting a source ranking.*

1. Introduction

Neural architecture search is bottlenecked by evaluation: training a candidate costs GPU-days, and a search visits thousands. Zero-cost proxies [1] aim to reduce this bottleneck, ranking an architecture from a single forward–backward pass at initialization, and on any *single* benchmark several rival a trained predictor [12]. They are often interpreted as *capacity-associated* surrogates, tracking parameters and FLOPs about as well as accuracy on some benchmarks [11, 17, 19], a proposed route to cheap evaluation.

However, two observations about this zoo are in tension: many proxies correlate with capacity, yet proxy quality is not reliably portable [12, 17, 19]. A proxy that tops one benchmark or axis can be anti-correlated with accuracy on another. The condition-number sum cond-sum, for example, runs positive ($+0.88$) as wiring varies and negative (-0.77) as width varies: the same published statistic, with opposite sign across axes (Fig. 1). We ask which statistical structures accompany this transport failure and whether it changes a budgeted search outcome.

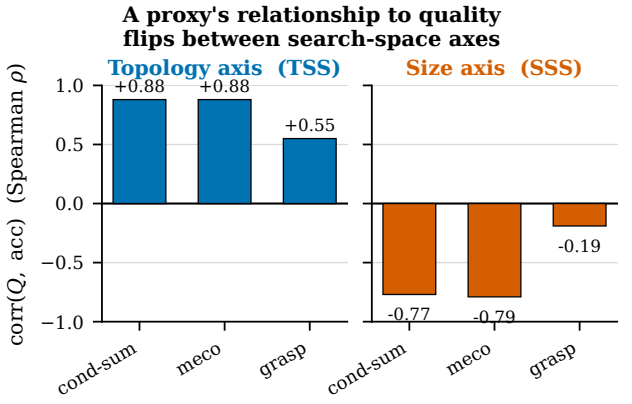


Figure 1. **The cross-space sign-flip.** A proxy's rank correlation with quality $\mathrm{corr}(Q, \mathrm{acc})$ reverses sign between the topology axis (TSS, *left*) and the size axis (SSS, *right*): cond-sum, meco, and grasp run $+0.88, +0.88, +0.55$ on topology but $-0.77, -0.79, -0.19$ on size (Sec. 5).

Rather than build a new proxy, we analyze the structure associated with existing scores. The clean instrument is NAS-Bench-201 [8]: its topology axis varies wiring at fixed width and its size axis varies width at fixed wiring, so each axis turns *one* knob, where a cross-*benchmark* comparison would confound family, dataset, and parameterization at once. Treating a 29-feature aligned zoo as one instrument, we organize each proxy's accuracy correlation (Eq. (1)) into a *capacity* term, computable label-free from the representation spectrum, and a *trainability residual* read against a measured target T, the early-descent rate of the real training curve. Coincidentally, NAS-Bench-Suite-Zero reports an

ensemble adds up to $+42\%$ over the best single proxy [12]; we do not chase that number; we test how much of a related complementarity gain remains after capacity adjustment (Sec. G.7). This decomposition is descriptive rather than causal: it separates a dominant capacity-associated factor from an early-trainability residual and then tests how both behave under transport.

Taken together, the evidence supports three bounded conclusions (Figs. 1 to 3). First, both axes contain a dominant capacity-associated factor, but outcome-informed orientation overstates cross-axis loading-direction agreement. Second, a capacity-orthogonal early-trainability association remains partly readable at initialization. Third, several leading raw proxies reverse sign under transport, while other proxies— including recent controls—remain positive on both axes. Concretely, we contribute:

1. A **capacity–trainability decomposition** (Eq. (1)): a dominant capacity-associated factor (Sec. 4, with the narrow label-free yardstick of Proposition 1) and a smaller, capacity-orthogonal trainability association (Sec. 4) that is partly init-accessible (Sec. 4).
2. An **empirical test** of a proposed "label-free ceiling" on these benchmarks (Sec. 4): on the topology axis the best proxies largely close the gap themselves, so the binding limitation there is portability, not magnitude; on the size axis the residual increment sits at the noise floor.
3. A **tiered portability map** (Sec. 5, Fig. 3): three motivating within-NB201 examples, $9/13$ descriptive reversals across the broader map, and four strictly supported cross-benchmark instances. Dimension controls narrow cond-sum to a dimension-linked/proxy-geometry-dependent failure rather than an independent mechanism.
4. A **controlled search consequence** (Sec. 5): two named traps cost $+5.40$ and $+1.97$ points at $K{=}10$, while two null directions cost zero. The target-calibrated screen is separated from target-EVAL scoring but is not fully label-free, and we do not claim universal superiority.
5. A **reusable artifact design** (Sec. 5.1): we construct a cross-space portability map from per-architecture proxy values and per-(proxy, space-pair) sign-stability annotations.

Our study is *observational*: we turn no controlled knobs and make no causal claims; limitations, including the weaker size-axis arm ($n{=}138$ full-zoo), are detailed in Sec. 6. The appendix supplies the dependency layer under each result, organized as a per-conclusion dependency map (Sec. B).

2. Related Work

The zero-cost NAS literature has largely asked *how to build a proxy that ranks better*; we instead study the statistical structure associated with existing proxy scores, organizing each proxy's quality correlation into a capacity term and a residual trainability term (Eq. (1)) and studying the size, accessibility, and sign-stability of the two (expanded in Sec. E).

Capacity and expressivity proxies score the richness an architecture can express at init (NEAR [9] via effective rank, Dextr [2] via a spectral condition number (its cond-sum we reimplement at parity), Zen-NAS [15] via Gaussian-perturbation response) but none prices *how much of every other proxy is largely the same number*; Sec. 4 shows a single component aligns closely with effective rank. **Trainability and gradient proxies** score how easily a net will train, from one-minibatch gradient/Jacobian/NTK statistics (ZiCo [14], TE-NAS [6], jacob-cov [16], GradSign [20], MeCo [10]); what is open is whether, with capacity held fixed, anything remains for them to read at init, which Sec. 4 supply as the second term of Eq. (1). **Ensembles and joint proxies** mix the two (AZ-NAS [13] ensembles four axes, InTrain [21] multiplies a capacity term by a gradient-health term, while SWAP-NAS [18] scores sample-wise activation patterns). Several evaluated constituents reverse between axes, whereas SWAP and AZ-NAS remain positive on both (Sec. 5); this mixed evidence motivates validating a recipe's orientation rather than assuming universal transport. L-SWAG [5] is treated as uncovered rather than assigned an approximated score because no official, licensed, auditable implementation was available for this study. **Benchmarks and the cross-space critique** document these effects without a single account: rankings are inconsistent across spaces and many proxies correlate with accuracy via parameter count [17, 19], simple features rival learned proxies [11], and NAS-Bench-Suite-Zero [12] reports combining 13 proxies adds up to $+42\%$ over the best single one. We do not chase the $+42\%$ (our same-direction gain is a conservative $+13.8\%$ (TSS) / $+1.0\%$ (SSS)) but *relate the observed gain to two associations*: a leading capacity-associated term (Sec. 4) and a residual associated with the external early-training target (Sec. 4); the cross-space inconsistency is summarized by the sign reversals of Sec. 5; against our own framing we refute a strict label-free ceiling (Sec. 4). A measurement-provenance inventory and compute accounting are provided in Sec. J.1. **Kernel spectral bias** we borrow only as a map: the picture of Canatar *et al*. [4] (and Bordelon–Pehlevan) reads a label-free statistic's reachable target as the at-init NTK's kernel-regression error, and our single formal statement (Proposition 1) bounds that *frozen* target, not trained accuracy. The observed proxy–trainability associations therefore do not follow from the proposition (Sec. 4). The proposition is motivation; the contribution is the empirical decomposition.

3. Setup: One Instrument, One Knob at a Time

The object of this paper is a reversal visible in Fig. 1: a zero-cost proxy that is among the leading predictors of accuracy on one axis of NAS-Bench-201 becomes one of the more *anti*-correlated on the other, sign-reversed, the within-benchmark face of the field's documented cross-benchmark inconsistency. To make that reversal a clean measurement we hold the network family, the dataset, the proxy code, and the accuracy table fixed and vary *one* architectural knob: wiring on one axis, width on the other. This section pins down each controlled piece and states plainly what we do and do not claim.

Benchmark and two axes. We use NAS-Bench-201 [7] via NATS-Bench [8]. Its value is that it exposes *two* architectural axes over the *same* dataset (CIFAR-10), each varying a single knob. The *topology axis* (TSS) varies cell wiring (which operation occupies each of the six edges of a fixed-width cell) so architectures differ in *connectivity* at roughly constant width. The *size axis* (SSS) varies the channel width of a fixed cell topology, so architectures differ in *capacity* (parameters and nominal channel width) at fixed wiring. Studying both is the point, not a robustness afterthought: because TSS and SSS each turn one knob, the same statistic, the same code, on the same family and dataset, can relate to accuracy in opposite ways depending on whether topology or width varies (Fig. 1, Sec. 5). A single-axis benchmark cannot show this: the reversal needs two clean knobs that vary topology and width in different ways. Accuracies are the CIFAR-10 test accuracies cached in NB201 (TSS hp=200, SSS hp=90), never trained by us.

The 29-feature aligned zoo, and sign orientation. We analyze **29** aligned at-init features in four groups (Tab. 8 lists every member). *Standard* (12) collects established metrics used by NAS-Bench-Suite-Zero [12] (including zico [14], meco [10], zen [15], and ntk [6]). *Spectral* (7) are functionals of the feature-map singular spectrum: the Dextr inverse-condition-number sum cond-sum and stable sum stable-sum [2], and the Rényi-α effective ranks R_α for $\alpha \in \{0.5, 2, 4, 8, 16\}$; R_1, the effective-rank capacity anchor from NEAR [9], is held out from the proxy PCA. *Label-free probes* (9, the Q-* family) comprise four NTK/Gram-spectrum summaries (log condition, log trace, effective rank, and log largest eigenvalue), ZiCo-style gradient variation and feature/gradient coherence under random versus $90°$/channel-rotated inputs, and a rotation-adaptivity score. And a single *label-using oracle* (1), Q-zico-oracle, consumes the real labels: it is held apart as an upper reference, flagged ("LABEL") wherever it appears, and excluded from every label-free claim. This 29-feature aligned zoo is distinct from the 13-proxy common set used for the seven-column portability map. Proxies disagree on which direction predicts higher accuracy, so the aggregate analysis (e.g. the PCA of Sec. 4) orients each proxy *within each axis* by the sign of its Spearman correlation with accuracy, falling back to the documented prior direction only when the data correlation is negligible (plain, $\rho = -0.01$ on TSS). Because this rule uses the outcome separately on each axis, we also report raw and globally fixed orientation. Column-sign reflection leaves PCA variance and score/capacity association unchanged but reduces the cross-axis loading cosine from 0.869 to 0.461; we therefore do not interpret the outcome-informed directional loading agreement as portable structure.

The decomposition lens. We read the aligned zoo one term at a time. For a proxy Q the rank relationship to accuracy is organized into a *capacity-associated* part, a dominant principal component whose scores track effective rank on both axes, and a *trainability residual*, whose per-proxy direction is tested for transport sensitivity:

$$\underbrace{\mathrm{corr}(Q, \mathrm{acc})}_{\text{proxy}} \rightsquigarrow \underbrace{\text{capacity}}_{\substack{\mathrm{PC1} \approx \text{erank} \\ \text{(associated)}}} \rightsquigarrow \underbrace{\text{trainability}}_{\substack{\text{partial}(Q, T \mid \text{cap}) \\ \text{(association)}}} . \quad (1)$$

This is descriptive bookkeeping, not a numeric identity: Eq. (1) names the two terms Sec. 4 reads off in turn, and the formal yardstick below prices only its capacity term.

The trainability target T (external, not a proxy). Separating a capacity component from a trainability component requires a ground-truth notion of how readily an architecture trains, one that does *not* come from a proxy, or the separation collapses into a self-consistency check. We therefore define the trainability target T as the *early-training learning rate* T_{slope}: the OLS slope of the *train*-accuracy curve over the first epochs (hp=12, epochs 0–4), read off the NATS-Bench training curves, so larger T means faster early learning. T is finite on every sampled architecture on each axis and carries a monotone relationship to accuracy in the premise check. Because T comes from training curves rather than proxies, partialling a proxy against T is a comparison against an *external* benchmark measurement, which makes the downstream residual operationally distinct from proxy self-consistency. The three objects this study leans on (the zero-cost proxy, the kernel picture, and T) are fixed in Sec. C.

Faithfulness precondition: the spectral arm is faithful. Before reading any cross-space behavior off the zoo, we validate our spectral proxy implementations against the official Dextr harness. cond-sum reproduces the inverse-condition-number statistic at parity and attains Spearman $+0.8868$ with accuracy on TSS, consistent with the published Dextr value ≈ 0.88 [2]. The TSS parity check makes a simple implementation mismatch less plausible, but it does not by itself validate a causal account of the SSS behavior. The full parity check, the hook-exception count, and

the data-quality flags are in Sec. H.2.

Populations and sample sizes. Coverage differs by proxy family, so each axis carries two principal sample sizes (Tab. 7, Sec. H.1). The *full-zoo join* intersects all four measurement tables and contains $n = 324$ TSS and $n = 138$ SSS rows before analysis-specific finite-value masks. The 29-feature PCA-complete population is $n = 317/138$; decomposition and per-proxy analyses state their own eligible counts. The *standard-proxy* population uses only the standard proxies, which reach more architectures, $n = 1000$ (TSS), $n = 425$ (SSS), with effective rank and T present on the full-zoo subset. The SSS full-zoo arm (n=138) is the binding sample-size constraint, flagged throughout as the weaker arm. Three proxies are computed on separate single-architecture infer.tiny nets and merged in; two (nwot, epe-nas) return a degenerate value on chance-level architectures and are listwise-dropped, a conservative choice detailed in Sec. H.1. The inclusion audit confirms that the 138 SSS records arise from the deterministic table intersection, but also finds a mild shift toward higher-accuracy/larger models; we retain this representativeness limitation.

An observational study, stated up front. We turn no controlled architectural knob and intervene on nothing: we read proxies, accuracies, and training curves off cached benchmark tables and correlate them. Every number in Secs. 4 and 5 is therefore *correlational*, and we make no causal claims. The full decomposition is confined to NB201's TSS and SSS, the two axes where both T and the spectral arm are available, with the weaker SSS arm (n=138 full-zoo) flagged throughout; the cross-space flip on $\mathrm{corr}(Q, \mathrm{acc})$ we additionally confirm generalizes across NB101, NB201, NB301, and TransNAS-Bench-101 (Sec. 5). Read each subsequent number as a description of these benchmarks, not as a controlled or causal finding. Full populations, the leakage-clean cross-validation, and the bootstrap protocol are in Sec. H; every body number traces to a logged source (Sec. J).

A formal yardstick (one knob has a ceiling). The kernel spectral-bias picture offers a narrow formal yardstick. Within that picture the spectrum is label-free whereas the target coefficients are not; this does not characterize every signal an at-init statistic may carry. The standard map [3, 4] (qualitative only; restated in Sec. D) has one certifiable consequence for a fixed target and information set.

Proposition 1 (Label-free resolution limit)**.** *Fix a probability space and let $\mathcal{G}$ be the sub-σ-algebra generated by the label-free at-init information. For a target Y with $0 < \mathrm{Var}(Y) < \infty$, define the* frozen-kernel resolution limit

$$R^2_{\mathrm{cap}} := \frac{\mathrm{Var}\big(\mathbb{E}[Y \mid \mathcal{G}]\big)}{\mathrm{Var}(Y)} \in [0, 1]. \tag{2}$$

Then for any $\mathcal{G}$-measurable statistic S with $0 < \mathrm{Var}(S) < \infty$,

$$\mathrm{Corr}^2(S, Y) \le R^2_{\mathrm{cap}}. \tag{3}$$

Scope. Eq. (3) bounds the squared *Pearson* correlation (not the rank ρ we report), with Y read as the *frozen at-init NTK* kernel-regression error, *never* trained accuracy, and acquires content only on a *restricted* information set (a finite feature dictionary), since the full $\mathcal{G}$ contains architecture identity and makes the bound vacuous. Proof in Sec. F; the spectral-bias map and the longer scope discussion are in Sec. D.

This is a label-free statistic's ruler: it bounds the *frozen at-init NTK* error, never *trained accuracy* (proof Sec. F; the spectral-bias map is Sec. D).

4. What Recurs: Capacity Association and an Accessible Trainability Residual

We read the oriented 29-proxy zoo (Sec. 3) one term of the decomposition lens Eq. (1) at a time: summarize the zoo by PCA, treat the leading component as an empirical *capacity* axis, and ask what survives once capacity $\{\mathrm{erank}, \mathrm{params}\}$ is held fixed against the directly measured *trainability* target T. A scope caveat governs everything below: Proposition 1 prices only the capacity term of Eq. (1), and only for the frozen at-init NTK target, not for the trained accuracy these findings are about (Sec. 3). Both axes are analyzed, with sample sizes stated per analysis: the joined full-zoo cohorts contain $324/138$ rows, whereas the 29-feature PCA-complete cohorts contain $317/138$. Each result traces to a body exhibit and corresponding appendix evidence (Sec. B).

Associated structure in the zoo, in brief (Fig. 2). To first order the proxy zoo contains a dominant *capacity-associated factor* (PC1 scores track effective rank and explain ∼half the variance on both axes), but a capacity-orthogonal early-training term (partial $\approx +0.6$) remains after capacity adjustment and is partly predictable at initialization; the axis-specific best label-free proxies (meco on TSS, synflow on SSS) close most of the observed gap, so no empirical ceiling is detected on these benchmarks.

We summarize above and elaborate next, in the order the findings earn.

Capacity dominance (a confirmatory replication anchor). To first order the proxies share a capacity-associated factor: the leading component scores strongly track effective rank on both axes (n=$317/138$ complete cases). That component carries about half the variance (52.9% TSS / 49.4% SSS; Tab. 11). Raw and globally fixed orientation leave these quantities unchanged, but reduce the cross-axis loading cosine from 0.869 to 0.461; thus the shared factor recurs, while its loading direction does

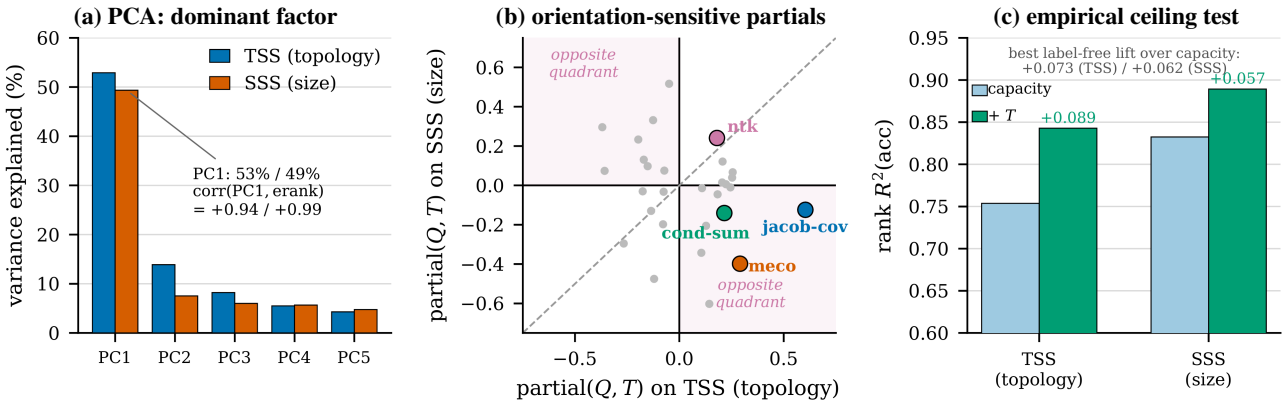


Figure 2. **Capacity is dominant; the early-training association is cross-space-fragile. (a)** PCA scree of the oriented zoo, PC1 tracking effective rank; **(b)** per-proxy partial partial$(Q, T \mid \text{erank}, \text{params})$ on TSS (x) vs SSS (y), opposite-sign quadrants shaded; **(c)** rank-R^2(acc) of capacity alone vs +ground-truth T vs +axis-specific best label-free proxy (meco/synflow).

not transport as strongly as the per-axis outcome orientation suggested (Fig. 2a; Supplementary Sec. I.1, *Orientation control*). This reproduces the established "proxies ≈ params/capacity" reading of [11, 17, 19], and serves as the capacity-association anchor for the rest of the section rather than a co-equal headline. The dominance is not an artifact of the Rényi sweep that supplies many oriented proxies: a 12-proxy rank-PCA with *no* Rényi sweep still leaves a single axis closely aligned with effective rank (Tab. 11). Exact PC1 correlations and the top-14 loadings are in Sec. G.6 (Tab. 4).

Matched-sample-count control. Across 1,000 without-replacement TSS subsamples of n=138, PC1 variance stays in $[0.497, 0.559]$, its effective-rank association in $[0.918, 0.953]$, and the cond-sum, meco, and grasp reversals retain their direction in every sample. These algorithmic summaries rule out simple sample-size imbalance, not the separate SSS representativeness limitation (matched-n intervals in Sec. I.2).

A capacity-orthogonal trainability association remains. A nonzero, capacity-orthogonal *trainability association* remains once capacity is fixed: partial$(T, \text{acc} \mid \text{erank}, \text{params}) \approx +0.6$, replicating across all three NB201 datasets. This term is what makes the decomposition worth running, since if capacity were the whole story nothing about trained accuracy would survive conditioning on it. The residual is carried by the externally measured early-training descent rate T (a benchmark measurement, not a proxy self-check): adding ground-truth T on top of capacity raises rank-R^2(acc) from 0.7537 to 0.8429 on TSS (95% CI $[0.798, 0.880]$) and comparably on SSS, increments that clearly exceed a random third predictor's noise floor (Fig. 2c) and stay positive under loss-based and slope-based T (Sec. H.3). Per dataset, parameters alone explain about half the rank-R^2 while T and effective rank each add a comparable capacity-orthogonal share, the two together reaching ~ 0.83 everywhere (Tab. 1; full grid Tab. 6).

	leakage-clean rank-R^2(acc) (R^2_{cv})			
Dataset	params	$+T$	+ erank	$+T$+ erank
CIFAR-10	+0.526	+0.775	+0.730	+0.828
CIFAR-100	+0.541	+0.743	+0.778	+0.840
IN16-120	+0.482	+0.777	+0.727	+0.837

Table 1. **The descriptive pattern recurs across all three NB201 datasets:** leakage-clean rank-R^2(acc) (n=324), where T and erank each add a comparable, capacity-orthogonal share on every dataset (Sec. 4).

Some early-training signal is predictable at init. This trainability residual is partly predictable from at-init statistics. Regressing capacity {erank, params} out of both accuracy and every proxy *inside each cross-validation train fold* (Sec. H.3), the recovered out-of-fold accuracy residual transfers to T itself (partial$(\widehat{\text{res}}, T \mid \text{erank}, \text{params}) = +0.536/+0.606$). The key number is what *survives* a strict leakage-clean test: the all-proxy-over-best-single complementarity gain, +0.2955 (+103.2%, TSS) / +0.0644 (+17.7%, SSS).[1] The held-out associations are strongest for jacob-cov (+0.604) and the Jacobian/NTK family on topology, the gradient proxies on size, most proxies not at all (Sec. G.7, Tab. 3). The per-proxy partial directions often differ under per-axis outcome-informed orientation (Fig. 2b), whose sensitivity is resolved in Sec. 5.

No empirical label-free ceiling on these benchmarks. The first three results invite a reading in which the rest of accuracy is a hard limit only training breaks, Proposition 1's

[1] The NBSZ-comparable pre-residual replicate is the smaller +13.8% / +1.0%; Sec. G.7.

ceiling reading transposed onto trained accuracy. The data do not support that reading on these benchmarks. Over capacity $\{\text{erank}, \text{params}\}$, meco on TSS and synflow on SSS already lift rank-$R^2(\text{acc})$ by $+0.0728$ / $+0.0616$, while the additional gap closed *only* by substituting ground-truth T is just $+0.0164$ (TSS) / -0.0048 (SSS) (Fig. 2c), too small to be a ceiling.

The direct NTK test preserves the proposition's narrow scope. On 324 TSS architectures, the frozen at-init NTK error correlates only ρ=0.40 $[0.29, 0.49]$ with trained accuracy, although a finite 30-feature dictionary predicts both well. Proposition 1 therefore prices that frozen-kernel target, not trained accuracy; full family and ridge results are in Sec. G.5.

For the evaluated examples, an early-training association is accessible at init, but its direction is less portable than its within-axis magnitude; this motivates the transport analysis in Sec. 5.

5. What Flips: The Cross-Space Sign Reversal

The portability failure, in brief (Fig. 3). A single zero-cost proxy reverses the sign of its relationship to quality when moved between two axes of the same benchmark. Three within-NB201 examples motivate the analysis; $9/13$ proxies reverse descriptively somewhere in the broader map and four cross-benchmark cases meet the strict support rule. The cond-sum failure is dimension-linked, and two named budget-search traps cost $+5.40$ and $+1.97$ accuracy points at K=10, while two null directions cost zero.

We summarize above and elaborate next.

The cleanest instance lives *inside* a single benchmark. The preceding analysis (Sec. 4) found a capacity-associated factor that recurs on both axes and a trainability residual that is statistically distinct and partly accessible. We now ask whether a single proxy carries the same signal once it is moved to a structurally different axis. NB201's two one-knob axes (Sec. 3) let a reversal be read off without blaming a change of dataset, recipe, or proxy implementation. We therefore center the orientation-free accuracy correlation $\text{corr}(Q, \text{acc})$ and treat the per-proxy trainability partial $\text{partial}(Q, T \mid \text{erank}, \text{params})$ as an orientation-sensitive secondary view.

The sign-flip. The cleanest, orientation-free result is the raw accuracy-slot flip: a single proxy's correlation with quality reverses from positive on the topology (TSS) axis of NAS-Bench-201 to negative on the size (SSS) axis, with no sign-orientation rule applied. The spectral conditioning functionals make this plain: cond-sum goes $+0.88 \to -0.77$ (n=324/138), meco $+0.88 \to -0.79$, and grasp $+0.55 \to -0.19$, the page-1 headline (Fig. 1). The spectral arm also clears a faithfulness check: the TSS value is consistent with Dextr's published ~ 0.88 (cond-sum = the official inverse-condition statistic, 0 hook exceptions; Sec. H.2), making a simple TSS implementation mismatch less plausible. A reversal this large in the raw, un-oriented accuracy slot is the central flip we analyze.

Under per-axis outcome-informed orientation, the same reversal also reaches the trainability partial $\text{partial}(Q, T \mid \text{erank}, \text{params})$, the very slot previewed in Fig. 2(b), though here the evidence is more graded. The CI-separated trainability-slot reversals across both axes are carried by meco ($+0.291 \to -0.398$) and Q-adapt-rot ($+0.143 \to -0.602$); the headline cond-sum SSS trainability partial is -0.141 with 95% CI $[-0.31, +0.04]$, which spans zero, so it is consistent-with rather than a CI-separated trainability-slot flip, as are the SSS partials for jacob-cov and gradsign (Sec. G.1, Tab. 3). One caveat is decisive: because each proxy was sign-oriented per axis by its own accuracy correlation, the trainability-slot direction is entangled with the outcome-informed rule. In the global-orientation control, PCA variance and score/capacity association are unchanged but loading agreement drops from 0.869 to 0.461; meco's SSS partial also changes orientation. We therefore treat these trainability-slot directions as orientation-sensitive exploratory evidence, not as an independent cross-axis flip. The raw accuracy-slot reversals above remain orientation-free.

The outcome-informed orientation's narrow survivor. Under per-axis outcome-informed orientation, one proxy—ntk—has a *trainability partial* $\text{partial}(Q, T \mid \text{erank}, \text{params})$ that is positive with a CI excluding zero on both axes ($+0.180$ TSS / $+0.241$ SSS); its T_{slope} partial also survives BH-FDR on both axes. The qualifier matters: ntk survives *in this trainability-partial slot at the full-zoo sample*, not as a portable accuracy predictor—as a raw accuracy ranker it is anti-correlated on *both* axes (Tab. 3). This is not an orientation-robust uniqueness claim: under the globally fixed rule both meco and ntk have positive marginal intervals on both axes, an exploratory, unadjusted comparison. At the larger standard-proxy base synflow and zico also join ntk under this rule; per-proxy cells are in Sec. G.2.

The cond-sum failure is dimension-linked, not an identified mechanism. On balanced n=500/500 cohorts, the raw correlation is $+0.887/-0.781$ (TSS/SSS). Dividing by cached spectrum length gives $+0.842/-0.785$, and dividing by effective rank gives $+0.418/-0.885$. Within SSS, however, the partial of the raw score is only -0.079 after effective-rank control and -0.068 after joint cached-length, effective-rank, and parameter controls. Exact cached-length matching is not identifiable because the supports do not overlap (TSS 142–502; SSS 118–122). An unmatched effective-rank common-range

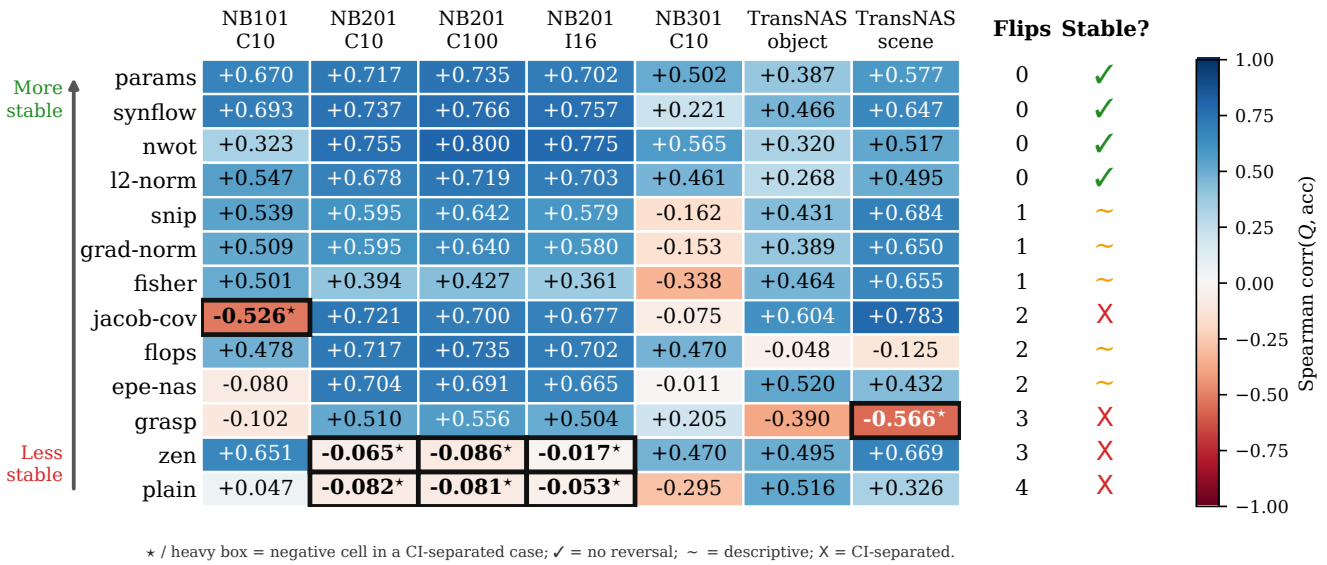


Figure 3. **Cross-space portability matrix.** Spearman corr(Q, acc) for 13 standard proxies over 7 search spaces, with rows ordered most-to-least stable (flip count ascending, then mean $|\rho|$). Heavy boxes and ⋆ mark negative cells in the four CI-separated cases (jacob-cov, grasp, zen, plain); the side panel reports each proxy's flip count and stability tier (✓: no reversal; ∼: descriptive reversal; X: CI-separated reversal). Nine of 13 proxies reverse descriptively on at least one space, while four meet the strict rule. Per-space sample sizes and intervals are in Tab. 5 and Sec. G.3.

sensitivity yields $-0.318/-0.885$, removing the cross-axis sign difference, but only 103 TSS records fall in that range. Finally, the cached count is not a validated representation dimension. These results support a dimension-linked/proxy-geometry-dependent transport failure. An exact representation-dimension-matched test cannot be performed because that field is absent from the available artifacts.

Score correlation (n=500 per axis)	TSS	SSS
cond-sum (raw)	+0.887	**−0.781**
cond-sum/cached length	+0.842	**−0.785**
cond-sum/effective rank	+0.418	**−0.885**
$R_{\alpha=1}$ (effective rank, NEAR)	+0.864	+0.911

Table 2. **Cached normalizations do not by themselves remove the full-population flip.** Effective-rank partial/common-range sensitivities and the non-overlapping cached-length support require the narrower dimension-linked interpretation in the text.

Full normalization intervals, within-length strata, the common-range sensitivity, and the reason exact dimension matching is unavailable are in Sec. I.3.

It partly generalizes. Of the 9 proxies that reverse the sign of corr(Q, acc) on at least one of the seven columns, exactly four survive a strict test—two columns whose 2000-resample 95% intervals are disjoint, each individually FDR-significant—and they fall into two magnitude tiers (Fig. 3). Two are large bilateral flips: jacob-cov (+0.78 TransNAS-scene vs. −0.53 NB101) and grasp (+0.56 NB201-CIFAR-100 vs. −0.57 TransNAS-scene). The other two are attenuations-to-near-zero rather than full reversals: zen (+0.67 vs. −0.086) and plain (+0.33 vs. −0.082), whose negative side is significant only by the large NB201 sample. The remaining candidates fail the bar—small-n NB301 attenuations, epe-nas's lone NB101 cell, and flops's near-zero TransNAS reversal—and only NB301 stays excluded as degenerate (Sec. G.3). This cross-space arm reintroduces the dataset and sample-size confounds the within-NB201 design removed, so these four flips are weaker evidence than the within-benchmark reversal. The decomposition behind this also replicates across NB201's three datasets—capacity stays the leading term and trainability a comparable contributor (Tab. 1)—while the deep capacity/trainability decomposition itself remains NB201-only. Full per-space values and CIs are in Sec. G.3, Tabs. 5 and 12, and Sec. H.5.

Recent controls bound the claim. SWAP [18] is positive on TSS/SSS (0.798/0.778), as is AZ-NAS [13] (0.922/0.740), with architecture-bootstrap intervals excluding zero. Thus SWAP is one sign-stable recent control, not evidence about every recent proxy; L-SWAG [5]/InTrain [21] remain unevaluated because no official, licensed, auditable implementation was available as of August 25, 2026 (intervals and protocol in Sec. I.6).

The full decomposition remains NB201-only. Across

49,674 audited external records, official artifacts expose only 13 finite proxy fields and lack the complete aligned proxy/capacity/training field classes required by Eq. (1). No population jointly has sufficient proxy coverage, effective rank, exact early T, and final accuracy, so Eq. (1) has zero eligible external rows. We report the missing-field coverage instead of constructing a surrogate, while retaining the raw map beyond NB201 (audit in Sec. I.5).

The flip has a search cost. The reversal changes which architecture a budget-K search returns. The baseline transfers the single best-on-source proxy as-is; the sign-stability screen uses a target-CALIB split to reject a proxy whose orientation does not transport, then scores a disjoint target-EVAL split. No network is trained, but cached target accuracy is used for calibration. On the TSS→SSS trap, the gap to the source-best proxy is $+7.83/+6.62/+5.40/+3.00$ at $K{=}1/5/10/50$. The scene→NB101 trap gives $+1.11/+1.55/+1.97/+1.56$; the reverse SSS→TSS and scene→NB201 nulls are zero at every budget. At the pre-specified moderate budget $K{=}10$, percentile intervals under the target-position resampling protocol are $[4.386, 6.435]$ and $[0.626, 3.133]$ for the two traps (Tab. 3). All $K{=}1/5/10/50$ results are retained to avoid favorable-budget selection. These are named trap costs, not average gains.

The diagnostic is target-calibrated, not fully label-free. The CALIB and EVAL positions are disjoint under the target-position resampling protocol, but canonical architecture identities can overlap. A second protocol partitions identities first and has zero CALIB/EVAL identity overlap; its K=10 intervals are $[5.057, 5.599]$ and $[1.487, 2.208]$, with both nulls still zero. The protocols use different resampling units and are reported separately. The evaluation also includes params, SynFlow, fixed rank-ensemble, random, negative, and unfavorable comparisons. The screen is useful for the two evaluated trap directions, but does not dominate every baseline and is not a universal NAS remedy. Full budget results, baseline comparisons, and the identity-disjoint sensitivity are reported in Supplementary Sec. I.4 (*Search consequence and sensitivity protocols*).

Table 3. **K=10 search consequence** (cached accuracy, no training). Values are target-CALIB screen minus transported source-best proxy; intervals are non-inferential percentile summaries under the target-position resampling protocol.

Direction	gap (pts)	percentile interval
TSS→SSS (trap)	+5.402	[4.386, 6.435]
SSS→TSS (null)	0.00	[0, 0]
scene→NB101 (trap)	+1.973	[0.626, 3.133]
scene→NB201 (null)	0.00	[0, 0]

Multiple-comparison control. The strict cases do not rely only on unadjusted screening: 34 of the 58 per-proxy partial cells and 78 of the 91 portability cells survive BH-FDR, and each of the four CI-separated flips is individually FDR-significant; the full accounting is in Tab. 9 (Sec. G.3).

5.1. A reusable cross-space portability map

We construct a $13{\times}7$ cross-space portability map (Fig. 3, Tab. 5) from per-architecture proxy values and per-(proxy, space-pair) *sign-stability flags*. These full-data flags are retrospective descriptive annotations, not inputs to the evaluated screen. In the search experiment of Sec. 5, the target sign is re-estimated separately within each split from target-CALIB cached accuracy, without using target-EVAL. A population-reconciliation table (Tab. 20) and the full datasheet (schema, splits, intended use, limitations; Sec. J.2) accompany it; the map will be released under a permissive open-source license, with a permanently archived and citable snapshot, upon publication.

6. Conclusion and Limitations

What travels, and what flips. The diagnostic picture is mixed. A dominant capacity-associated factor recurs on both axes: its variance share and score/effective-rank association recur, although its loading direction is outcome-orientation-sensitive (Sec. 4). Raw proxy/accuracy relations can *flip* (e.g. cond-sum $+0.88 \rightarrow -0.77$; Fig. 1). A capacity-orthogonal early-trainability association remains partly accessible at initialization, although its per-proxy directions are also orientation-sensitive (Sec. 4). Thus cond-sum indicates a dimension-linked/proxy-geometry-dependent transport failure, not an identified mechanism. Our data also do not support a "label-free ceiling": meco on TSS and synflow on SSS recover much of the trainability gap (Fig. 2c; Sec. 4). Proposition 1 certifies only a squared-correlation bound for a *frozen at-init NTK* target, not trained accuracy.

The practical rule. Validate a zero-cost proxy's orientation on the target axis before transporting its source ranking. In two named traps, a target-CALIB screen avoids $+5.40$ and $+1.97$ point costs at $K{=}10$; two null directions remain unchanged (Sec. 5). The diagnostic uses cached calibration labels, does not dominate every baseline, and is not a fully label-free or universal remedy.

Limitations. All results are observational benchmark correlations. Matched-n resampling addresses sample count, but SSS representativeness and true dimension matching remain unresolved. External flips reintroduce dataset/n confounds; the decomposition is NB201-only, and unconstrained deployment plus optimizer/initialization/regime interventions remain future work.

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